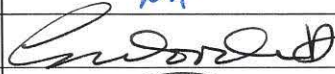


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☐ Vendor Study Protocol		☐ Vendor Study Report		☒ Lpath Study Report	
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Executive summary of study results: ☐ See Attached ☐ N/A Lpath has developed murine and humanized monoclonal antibodies targeting LPA. This report covers development of ELISA methods that are both sensitive and specific for Acyl chain LPA isoforms. The Ki of both the murine and humanized antibodies range from 15.8 nM to 986nM for all of the isoforms tested					

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Direct and Competitive Binding of anti-LPA antibodies B3 and LT3114

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1 SUMMARY

Lpath has developed murine and humanized monoclonal antibodies targeting LPA. This report covers development of ELISA methods that are both sensitive and specific for Acyl chain LPA isoforms. The K_i of both the murine and humanized antibodies range from 15.8 nM to 986nM for all of the isoforms tested

2 INTRODUCTION

Lysophosphatidic acid (LPA) is not a single molecular entity but a collection of structural analogs with a single functional glycerol alcohol phosphate moiety esterified to a variety of fatty acyl hydrocarbons of varied lengths and degrees of saturation. LPA species with both saturated (16:0 and 18:0) and unsaturated fatty acids (16:1, 18:1, 18:2, and 20:4) have been identified in biological fluids with 16:0, 18:1 18:2 and 20:4 LPA being the predominant species. Lpath has generated murine and humanized monoclonal antibodies to lysophosphatidic acid and this report describes data showing that these antibodies bind to all of the relevant isoforms of LPA.

3 MATERIALS AND METHODS

3.1 Test Article and Control

Murine anti-LPA antibody (504B3; B3) Lot # HR121456

Humanized anti-LPA antibody (LT3114) Lot # LP121425

3.2 Study Design

Development of both direct and competition ELISA format assays to demonstrate binding and specificity of anti-LPA antibodies to LPA ligands.

3.3 Direct Binding ELISA

The assay uses thiolated C12-LPA conjugated to BSA as the capture agent for the direct format ELISA. Ninety six well plates were coated with 100 μ l of 1 μ g/ml of C12 LPA-BSA-SMCC for 1 hour at 37 degrees. The plates were washed 4 times with PBS and blocked with 150 μ l of PBS-BSA-Tween for 1 hour at room temperature. 100 μ l of B3 or 3114 antibody was added starting at 4 μ g/ml and diluted 2-fold across the plate for the standard curve. Plates were incubated at room temperature for 1 hour. Next, plates were washed 4 times with PBS and 100 μ l of a peroxidase conjugated anti-mouse IgG (H+L) (Jackson immunoresearch 115-035-062) or anti-human IgG (H+L) (Jackson immunoresearch 709-035-149) was added at a 1:40K dilution and incubated for 1 hour at room temperature. Finally, the plates were washed four times with PBS and developed with 100 μ l cold TMB substrate for 12 minutes at room temperature. The reaction was stopped with 100 μ l of 1M H₂SO₄. Plates were read at 450 nm.

3.4 Competition ELISA

The specificity and binding of the murine (B3) and humanized (LT3114) anti-LPA monoclonal antibodies were determined by competition ELISA. For LT3114 competition, goat anti-human IgG (Jackson ImmunoResearch, Cat#109-005-098, Lot # 90887) was diluted to 1 μ g/mL in carbonate buffer (100 mM NaHCO₃, 33.6 mM Na₂CO₃, pH 9.5) and 15 μ L/well of the coating solution was added to 384 well plates and incubated overnight at 4°C. The following day, the plates were blocked with 60 μ L/well PBS + 1% BSA + 0.1% Tween-20 for 1 hour at room temperature. The human 3114 antibody was diluted to 50 ng/mL in PBS +0.1 % Tween-20 and 15 μ L/well were added and incubated for 1 hour at room temperature. Then, the plates were washed 4X with PBS and incubated for 2 hour at room temperature with 15 μ L/well LPA-biotin (20 nM) /unlabeled (2000 to 0.1 nM) lipid mixture. Next, plates were washed 4X with PBS and 15 μ L/well SA-HPX 1/30,000 was added and incubated for 15 min at room temperature. Again the plates were washed 4X with PBS and 15 μ L/well cold TMB substrate was added. After 5 minutes, the reaction was stopped with 1M H₂SO₄. The optical density (OD) was measured at 450 nm using a Thermo

Multiskan EX. Raw data were transferred to GraphPad software for analysis. Values for percent inhibition were calculated for each lipid using Microsoft Excel.

For B3 competition the same method was followed but Jackson Immunoresearch goat anti-mouse IgG, (Cat#115-005-071, Lot# 91365) was used to coat the plates.

3.5 Data Analysis

The plates were read on Thermo multiscan EX plate reader at an absorbance of 450 nm. Raw data was recorded directly by Ascent software for the Multiscan plate reader. Preliminary calculations and statistical analysis were performed either in Graphpad Prism or Excel software.

4 RESULTS

Binding of Anti-LPA antibodies to LPA by Direct ELISA

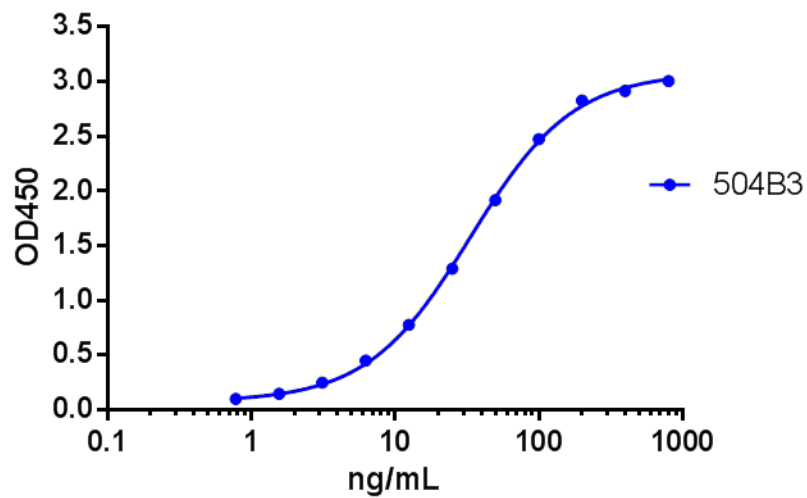
The murine mAb used for proof-of-concept studies in animal models of neuropathic pain (designated B3) and the humanized variant (LT3114) shows similar binding characteristics by direct ELISA. Binding curves were comparable and all fit well to the 4-parameter model with a Hill slope of ~1 and narrow 95% confidence limits for comparable EC50 (33.5 ng/ml for B3 and 24.4 ng/ml for LT3114) (Figure 1, A and B).

Binding Affinities of Anti-LPA antibodies to LPA Isoforms by Competition ELISA

The competition binding ELISA was used to compare the affinity of the murine (B3) and humanized (LT3114) antibody for different isoforms of LPA. Figure 2 shows the competition binding curves for 16:0, 18:0, 18:1, 18:2 and 20:4 LPA isoforms. The humanized antibody, LT3114, and the murine antibody, B3, from which the LT3114 CDRs were obtained, have essentially identical binding characteristics for native lipid, with a K_i of 16 nM for 18:2 LPA, 360 nM for 18:1 LPA (Table 1).

Figure 1. Binding Curves For B3 (A) and LT3114 (B)

A



B

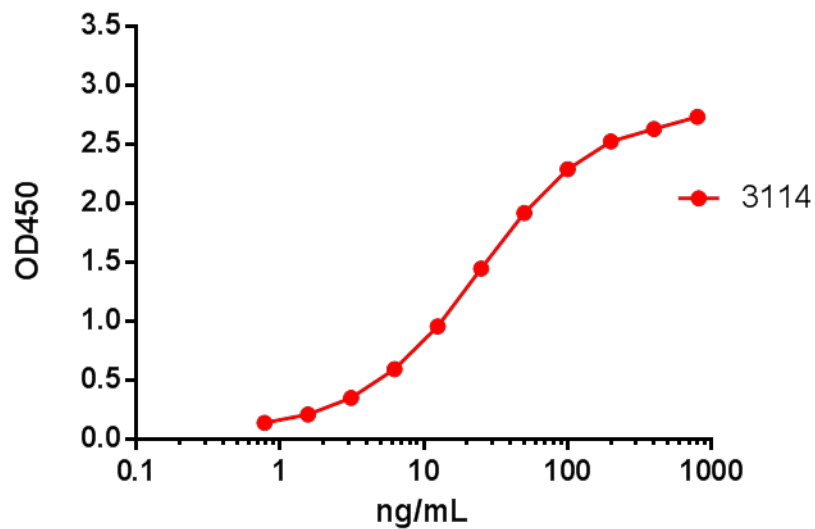


Figure 2 Affinity of anti-LPA antibodies for various LPA isoforms. Affinities of the antibodies for native 16:0, 18:0, 18:1, 18:2, and 20:4 were measured in an ELISA-based competition binding assay.

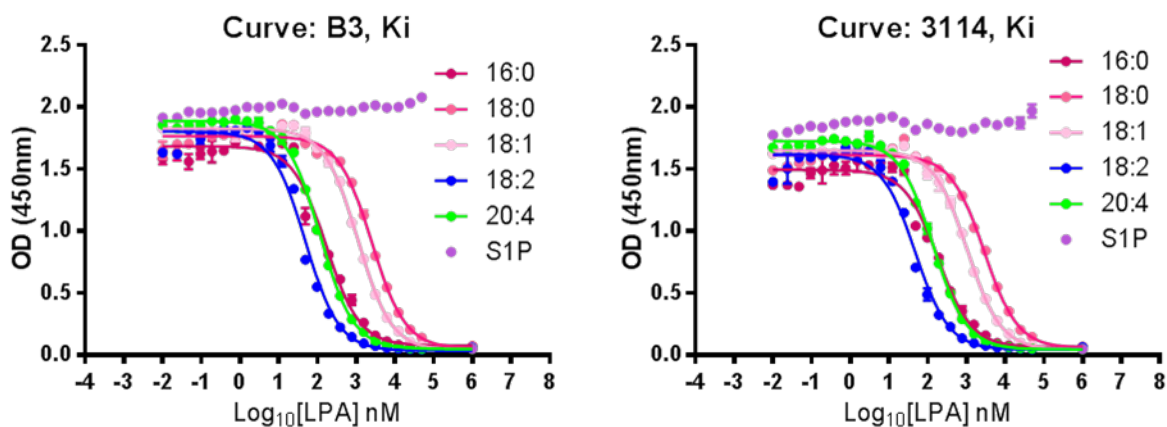


Table 1 The Ki for each antibody binding to native LPA isoforms.

LPA Isoform	<i>K_i, nM</i>	
	B3	3114
16:0	54.8	63.31
18:0	823.7	985.9
18:1	358.5	359.8
18:2	15.82	16.35
20:4	37.8	47.02

K_i for each of the lipids was determined by fitting the competition binding data using the formula of Cheng and Prusoff in GraphPad Prism software and the K_d of each antibody for the biotin-labeled LPA determined in a separate saturation binding ELISA experiment

5 CONCLUSION

We have generated potent and specific mouse (B3) and humanized (LT3114) neutralizing antibodies that bind to a wide range of physiologically relevant LPA isoforms.

6 DATA ARCHIVE

B3 Binding direct ELISA: Lpath Notebook #312, pp 5-6

3114 Binding by direct ELISA: Lpath Notebook #301, pp 102-103

Affinity data: Lpath Notebook #220, p81